

COURTSIDE: Calling the Point Before It Lands

Predictive Ball Tracking and the Value of Speed in In-Play Tennis Markets

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Abstract. We built a predictive ball-tracking engine that calls a point (ball out, or into the net) before the ball lands: on held-out table tennis it called 4 of 41 misses a median 162 ms early, none wrong, at 119.9 fps, and on broadcast tennis its out calls were right 5 of 5 at the bounce and 3 of 3 when 33 ms ahead. On 13,084 Polymarket ATP/WTB matches, wallets that trade within 3 s of a point earn after fees in all 11 months. Copying them at their own fills (v2) has a Sharpe ratio of 14.5 in sample and 6.7 out of sample (many small, nearly independent bets; Section 1); doubled fees erase it out of sample. A simulated computer-vision (CV) trader with an assumed 1 s feed makes +\$4 a day out of sample pre-registered and +\$57 (Sharpe 8.8) post hoc; with a licensed 0.5 s feed, +\$15 or +\$81, and on 16 newer matches pre-registered before we fetched them, −\$20.5 or +\$50.8 (anecdotal). Doubled fees and every blind test of a tradable book failed.

JEL: G14, G12, C58. **Keywords:** computer vision; ball tracking; prediction markets; latency arbitrage; in-play betting; tennis.

1 Introduction and summary

A camera films the court; a model finds the ball, fits its flight and calls the point before the bounce (Fig. 1). A Polymarket price is the chance that a player wins and lags each point by about a second, so whoever learns the point first sets the price. We map who is fast, price each second of feed delay (\$42–74 a day for the first) and test it on a pre-registered fresh holdout.

Our Sharpe ratios are above 3, so we hunted for bugs. v2's 14.5 comes from about 270 small bets a day on points already played, held to resolution, with a mean same-day correlation of 0.0013, so daily P&L is rarely negative; the CV trader is the same kind of event book. We found one bug: the score-move detector's start time used later prices (D9); the fix cut the in-sample Sharpe from 16.8 to 14.5. Then: at zero feed delay the latency sweep reproduces the published CV result exactly (the $V = 0$ identity); copying the same trades 3 s later loses (−1.05¢ a share IS), as does a copier filled after the 1.98 s block lag and the venue's hold (−0.47¢ IS, −1.37¢ OOS); our causal engine, with no look-ahead feature, nearly matches Table 1's 1 s cells; and 62 of 62 paper numbers re-derive in SQL (20 recomputed from rows, 42 looked up).

2 Economic hypothesis

A Markov model of points, games and sets gives the exact chance that each player wins from any score (Klaassen and Magnus, 2001), so each point moves fair value by a known amount at a known moment (the *swing*: mean 5.7%, 38.4% on a big point). The news reaches traders in tiers: court, video, then score feeds 28.2–44.1 s after the price moved. A maker whose quote still rests after the point is picked off by whoever knows first (Glosten and Milgrom, 1985; Budish et al., 2015; Aquilina et al., 2022), as courtsiders do in betting markets (Croxson and Reade, 2014; Brown, 2012): on 482 live points a median

Rerun: bash reproduce.sh (<https://github.com/OjasMishra32/computervisionpredictivetabletennismoneyonemachineouuushiii>). Prices, fills, fees and venue delays are real Polymarket records; only the CV trader's call timing, at an assumed feed latency, is simulated. Paper trading only: no real money, no licensed feed, no court camera.

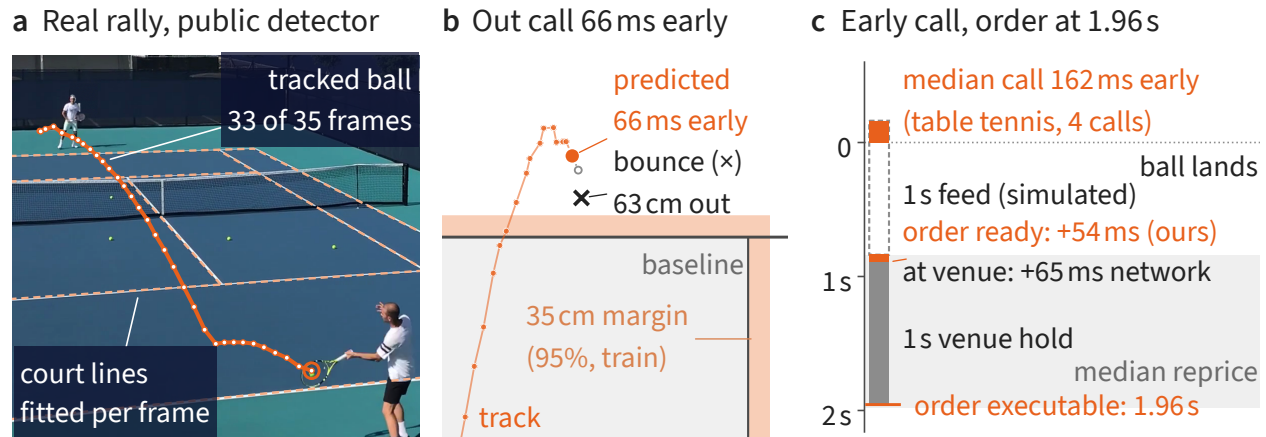


Figure 1. Our vision work: three separate pieces, not one run. (a) Licensed real rally: public TrackNet detector, our linking and court-line fit; ball in 33 of 35 frames. (b) Broadcast test bounce (the detector saw all ten games): out call 66 ms early at an assumed 25 fps (earliest of 3 of 3 right calls at that slot); landing 69 cm off, ball (x) 63 cm out. (c) Eq. 1: the order is 1,116 ms after the median reprice under the pre-registered stamp lag, 24 ms before it post hoc.

\$222 of quotes still rested at stale prices 0.25 s before the reprice (\$0 after 0.5 s), and Polymarket holds each taker order 1 s.

H6 (pre-registered). We expect wallets that trade within 3 s of a score move in Polymarket ATP and WTA match markets to earn a positive net *markout* over 30 s (profit at the price 30 s later, after fees), because the other side is a maker who has not yet cancelled a stale quote; the edge persists because physics and data licences decide who learns first. Wallets picked on past months then stay positive every month and out of sample; H6 fails if the markout is ≤ 0 in over a third of months, or out of sample. H1–H4 were committed before any result (7232986); H6, written after the in-sample tier results (D5), was frozen before any held-out read.

Will the edge last? The fast tier’s net edge fell 0.16¢ a month (CI 0.06–0.25), mostly as the fee rose; before fees there is no clear trend (–0.02¢, CI [–0.12, 0.07]). A line hits zero 3.3 months after September (CI 0.6–12.8), but a May step (picked by the data) fits as well with no trend: we cannot tell if the edge is dying, though a fee above 8.4% (today 5%) would end it, and entrants split the pool (one wallet’s monthly share fell from \$1,632 to \$387).

3 Data and universe

Every resolved ATP/WTA singles match market on Polymarket with at least \$5,000 traded: 13,084 matches (1,310 at Grand Slams; \$2.84B), Oct 2025–Oct 2026, from the public APIs (Polymarket, 2026a), point in time, so nothing is dropped for how it ended: 378 settled 50/50 (302 never started) and 283 ended in a retirement, all at their settlement price. *In sample* (IS) is the first 80% of matches by start time, *out of sample* (OOS) the last 2,617, from 2026-08-25 14:15 UTC (the 20%-or-two-years rule); we looked at it while building v2 and the CV simulation, so it is *burned*. Blind tests used 11,307 held-back markets and 16 newer matches; vision and point data are in Table A2 (Huang et al., 2019; Voeikov et al., 2020; Sackmann, 2026).

Data problems. Six tapes hit the API’s row cap (kept); 2,889 early matches had no published fee schedule (charged none); tape times are block times 1.98 s late, so signals use only prints stamped before them; gaps are carried forward, never filled from later prints (the CV trade set, selected on outcomes, is the labelled exception); no corporate actions apply. Regimes: the taker hold fell from 3 s to 1 s on May 15 and the fee rose from 3% to 5%; v2 is positive in each (+1.96¢ to +1.02¢ IS), and the CV trader is simulated only on the 7,755 matches with the 1 s hold.

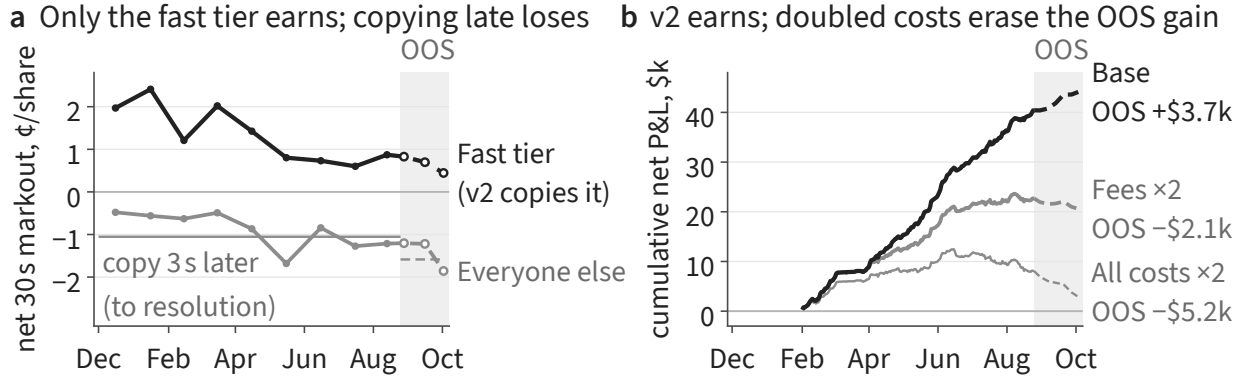


Figure 2. *Only the fast tier earns, and the edge is thin.* (a) Monthly net 30 s markout (copying 3 s later: held to resolution). (b) v2's equity curve, base and doubled costs. Source: results/v2.

4 Methodology

The fast tier and v2. Each month a wallet joins the fast tier if, in earlier months, it traded at least 30 times 0–3 s after a score move (a $\geq 4\text{¢}$ move of the 10 s price from its prior 60 s) in 10+ matches ($t > 3$). v2 copies their trades, sized by resolution risk (best of 55 rules, walk-forward), with \$1,000 orders and 100 shares net per match, held to resolution. Copying is not new; we add picking wallets by speed relative to the point, the sizing and limits, and the test that a copier loses: v2 prices speed and is not our trade.

Being fast ourselves. A CV model watches the video, calls the point early and sends a taker order before the stale quote is cancelled. Without video of the matches we trade, we simulate it on past points with an assumed feed latency (licensed feed not purchased); all other parameters are measured. The order wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (1)$$

with V the feed delay, δ our pipeline plus network, D the 1 s hold, ℓ how early the CV calls, L the *stamp lag* (landing to official stamp) and R the reprice relative to the stamp (median -1.16 s). Nobody publishes L : we *pre-registered* 2.0 s and also show 3.14 s, a *post hoc* estimate from one day. The trades are past points where the price later moved $\geq 4\text{¢}$: *selected on outcomes, not ex ante*.

Vision and speed. On held-out TrackNet broadcast games one TV camera's out calls were right 5 of 5, 3 of 3 and 3 of 4 at 0, 33 and 67 ms ahead, but only bounce-time calls met our 95% rule. Our live engine makes about 30 phantom calls an hour outside play (8 with a rally gate): not yet safe to trade. With a simulated 1 s feed, our 54 ms pipeline, 65 ms of network and the hold, an order can execute 2,119 ms after the point (p99 2,218 ms; built, not sent), inside the 3 s bar. A teammate's London server saw the book first on 99.74% of 287,577 messages (clock-corrected); with measured cold REST round trips (70 ms London, 184 ms Florida) the order executes in 2,089–2,146 ms (2,512 ms at worst; not order latency).

Costs and validation. Each match charges its own taker fee $f(q) = r q(1 - q)$ a share, 10,000 $r(1 - q)$ bps of notional (Polymarket, 2026a, fee schedule); today $r = 5\%$, 250 bps at 50¢. Fees paid were 120 and 179 bps of notional for v2 (IS, OOS) and 163–226 bps for the CV trader; v2's real fills include the spread crossed, and the CV fill walks the measured stale book. The stress doubles the fee (v2: also the spread). We tune only in sample, walk-forward.

5 Results

The copy book and overfitting checks. The fast tier earns every month and everyone else loses (Fig. 2a); v2 earns in both periods, thinly. Six tests on daily P&L (PSR and minimum track record against Sharpe 0 and 2, Bailey and López de Prado, 2012; a Bonferroni haircut over all 4,219 variants, Harvey and Liu, 2015; a 5-day block bootstrap): v2 passes all six in sample (haircut Sharpe 13.4) but

Table 1. Every book whose Sharpe ratio we print, net of costs: v2 (our copy of the fast tier at its real fills) holds out of sample but thinly; the simulated CV trader needs a fast feed and the post hoc stamp lag; doubled fees erase both out of sample.

	\$ a day	¢ a share	Return	Volatility	Sharpe	Max DD	Turnover	¢, fees ×2
v2, IS	+\$196	+1.38	253%	17.5%	14.5	-2.0%	93	+0.77
v2, OOS (burned)	+\$92	+0.60	148%	22.2%	6.7	-2.1%	129	-0.34
CV trader, pre-registered $L = 2.0$ s, simulated; break-even V: 1.09 s IS, 1.01 s OOS								
0.5 s, IS	+\$28	+0.61	51%	11.9%	4.0	-4.5%	36	-0.20
0.5 s, OOS	+\$15	-0.02†	32%	12.3%	2.1	-3.8%	43	-1.04
1 s, IS	+\$15	+0.40†	30%	12.9%	2.1	-6.6%	31	-0.41
1 s, OOS	+\$4	-0.38†	11%	15.2%	0.3	-5.3%	38	-1.40
3 s, IS	-\$13	-1.35†	-57%	24.3%	-2.6	-24.8%	25	-2.14
3 s, OOS	-\$13	-1.69†	-98%	42.0%	-2.6	-19.0%	44	-2.71
CV trader, post hoc $L = 3.14$ s (CI 2.23–3.22 s); break-even V: 2.23 s IS, 2.14 s OOS								
0.5 s, IS	+\$133	+1.21	146%	9.4%	15.5	-1.3%	59	+0.41
0.5 s, OOS	+\$81	+0.82	123%	9.9%	12.2	-1.3%	68	-0.21
1 s, IS	+\$94	+1.11	118%	9.8%	11.9	-1.6%	51	+0.32
1 s, OOS	+\$57	+0.66	91%	10.1%	8.8	-1.6%	59	-0.37
3 s, IS	-\$6	-0.79†	-18%	16.8%	-1.4	-14.4%	23	-1.58
3 s, OOS	-\$11	-1.47†	-66%	31.7%	-2.4	-13.9%	37	-2.48

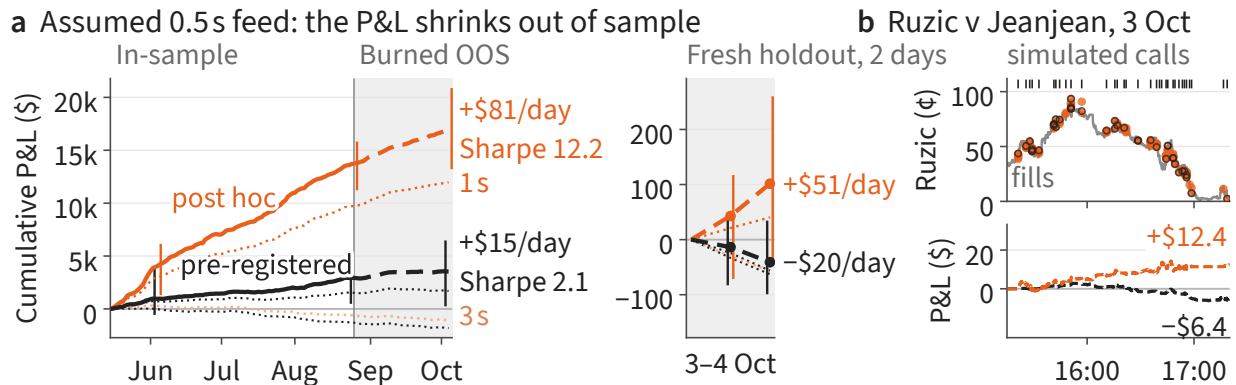
Notes: return and volatility a year on capital (3× peak dollars locked); max DD on capital; turnover: dollars traded ÷ capital a year. v2: ¢ CI [1.17, 1.59], [0.09, 1.13]; Sharpe CI [11.9, 17.4], [1.9, 12.3]; deflated Sharpe 0.997, 0.075 (3,386 trials, our count then); skew +0.53, +0.33; worst month +\$2,543, -\$434; all costs ×2: +0.27, -0.84¢. CV: assumed feed latency (licensed feed not purchased); parameters measured; 20 seeds; †: ¢ CI spans zero.

3 on the 40-day burned OOS (PSR vs 2 is 0.94, MinTRL to beat 2 is 44 days, the haircut Sharpe is 0: $t = 2.21$ vs 4.38). The pre-registered CV trader fails all six, the post hoc passes 4 OOS: none passes all six out of sample (even an optimistic copier makes only +0.21¢ [-0.32, 0.73] OOS). Nearby settings work: all 55 sizing rules have an in-sample per-share CI above zero; PBO is 9–24%.

What speed is worth. Pre-registered, the CV trader breaks even at a 1.01 s feed delay out of sample. Post hoc it makes +\$57 a day at 1 s, but read per point the post hoc inference loses (-\$17 a day). Every reading loses at 3 s, doubled fees make every OOS cell lose (best -\$13 a day), and a replay of 9 live-recorded matches calling every point loses in 36 of 36 settings: the gap is knowing which points matter.

If we were a quant firm with a licensed 0.5 s feed (Fig. 3). Counterfactual: real prices, fills, fees and hold; simulated calls; no feed bought, no order placed. At $V = 0.5$ s the Table 1 trader makes +\$28 / +\$133 a day IS and +\$15 / +\$81 on the burned OOS (pre-registered / post hoc). Before fetching newer data we pre-registered a fresh holdout (dc95717): the 16 covered matches that started after Oct 3, 07:10 UTC and had finished (2 UTC days; the 9 that built the model left out). It returned -\$20.5 / +\$50.8 a day (CIs [-\$96, +\$57], [-\$38, +\$141]; 7–10 traded matches), at 1 s -\$28.1 / +\$20.2, at 3 s -\$31.4 / -\$28.3: anecdotal, evidence neither for nor against an edge. Break-even feed price a month: \$787 / \$3,976 (IS), \$384 / \$2,397 (burned OOS) and -\$701 / +\$1,467 (fresh), against quotes of \$1,250–\$10,000: only post hoc beats the lowest. The showcase, fixed in advance as the covered match with most in-play volume (Antonia Ruzic v Leolia Jeanjean, \$190k), made -\$6.4 / +\$12.4.

Blind tests. On 11,307 never-examined markets (pre-registered) frozen v2 earned +2.02¢ IS but +1.22¢ [-0.19, 2.65] OOS: it fails, as did the maker book and rule v3 (Appendix B). Over 4,219 variants and 111 held-out reads we changed rules after a look 2 times for v2 and 13 for the CV simulation.



Counterfactual. Real Polymarket prices, fills, fees and venue delay; simulated camera calls at an assumed licensed-feed delay (no feed purchased).

Figure 3. A firm with a licensed 0.5 s feed (counterfactual). (a) Equity curves of Table 1's CV trader (20-seed mean, whiskers 10–90%; black pre-registered, orange post hoc, dotted 1 s and 3 s). (b) The pre-registered showcase match. Trades selected on outcomes.

6 Risk management

Limits and the most we can lose. Orders are at most \$1,000 and a match at most 100 shares net at 5–95¢, which caps the open position at \$95, not the loss: v2 buys both tokens in 79% of matches IS (71% OOS), and the worst match made –\$206 IS, –\$154 OOS. Only the backtest's \$3,000 gross cap per match bounds a match, at \$1,479; the live engine lacks it. A \$1,000 daily loss ends the day. Set in advance, we stop on a stale feed or vision, or slow latency, halve size if the trailing 30-day edge falls below 0.3¢, and stop at zero, at a 5% drawdown or a fee or hold change. None fired in sample.

Model and concentration. Wrong CV calls always fill, so out calls need confidence ≥ 0.95 and one false call halts the day. The top five copied wallets carry 81% of v2's P&L IS and 142% OOS. A per-wallet limit (\$1,000 a day, retiring wallets whose trailing edge is ≤ 0 ; backtest only), pre-registered on IS, kept 63% of IS P&L and doubled its drawdown; once on the burned OOS it made about the same (\$3,615 vs \$3,688; Sharpe 7.6 vs 6.7, CIs overlap), with a better worst day but worse drawdown, and the top five still at 129% (the rest lose at the same rate): a loss limit, not a fix. Settlement, US access and ticket terms add risk (Polymarket, 2026c,b; International Tennis Federation, 2026); factor risk is nil (largest $|t| = 1.36$).

7 Liquidity and capacity

v2 holds its OOS edge up to \$22,754 of capital (5× losses), trading 0.07% of match volume; the CV book's Sharpe halves at \$40,000–\$73,000 post hoc (Fig. A6), and pre-registered it has no capacity. Price impact, fitted on 4.5 million IS taker orders with a usable mid, grows with the log of size (0.75¢ a share at 1–10 shares, 1.17¢ at 10,000+ in the highest-volume fifth), the mid moves 0.05–0.14¢ more per 1,000 shares at 30 s (less on burned-OOS tapes: the fit is unstable), and in four hours of live book, buying just after a reprice cost 1.71 times a quiet book. With that charged, v2's Sharpe still halves near \$28.0k OOS, but 1× earns \$88 a day, not \$92 (\$71 conservatively, interval [–\$7, \$150]); CV capacity moves under 1%. Taking half the stale depth is cautious: in 907 live reprices and 265 stamped points, other takers took 4–11% of it. COURTSIDE prices speed; it is not yet a business.

8 Limitations and next steps

The stamp lag L is unmeasured and moves the break-even one for one; trading is simulated, the vision is tested on other matches and the fresh holdout is two days. A fee rise, a shorter hold or more fast wallets would break the edge. Next: a week of licensed feed to measure L , the forward test, and the limits in the live engine.

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Disclosure. We used no real money and connected no funded account. Simulated parts are labelled where they appear. We hold no licensed data feed, no footage of the matches we trade and no court access. Public datasets we did not build: the Match Charting Project (CC BY-NC-SA 4.0), the TrackNet broadcast tennis set and the OpenTTGames dataset. Open-source parts we did not build: the BlurBall and WASB model weights (Gossard et al., 2026; Tarashima et al., 2023), and the TrackNet weights and TennisProject/TennisCourtDetector code (yastrebkvs, 2026) (used unmodified, for evaluation only). The tennis rally is a Pexels stock clip (Gelato Prod, 2021); the logo on page 1 is the Gator Quant Hacks 2026 event's own mark. Authors: Ojasva Mishra, Yoan Exposito, Rafael Penhas, Ian Hoang (University of Florida). Teammates' work used here: Yoan Exposito ran the London latency probe on a Vultr server (sponsors/vultr), Ian Hoang re-derived 62 of 62 of our numbers in SQL on Snowflake and DuckDB, 20 recomputed from rows and 42 looked up (sponsors/snowflake), and Rafael Penhas built a Tiger Data tick store and live lab on the same tapes (sponsors/tigerdata).

Appendix A Calculations

Everything needed for scoring is in the main pages. Below: formulas with worked examples (A), every test (B), extra figures (C) and the records (D). Each value is in results/paper/numbers.json with its source file.

A1. Sharpe ratio. $SR = \sqrt{365} \bar{d} / s_d$, where d_t is net P&L on every calendar day (zero on days without trades) and s_d its standard deviation. v2 IS: $\bar{d} = \$196.24$, $s_d = \$258.87$ over $T = 206$ days, so the daily ratio is 0.758 and $SR = 0.758 \times 19.105 = 14.48$. OOS: $\$92.19 / \$264.15 \times 19.105 = 6.67$.

A2. Sortino ratio. $\sqrt{365} \bar{d} / DD$, where the downside deviation is $DD = \sqrt{\frac{1}{T} \sum_t \min(d_t, 0)^2}$. For v2 OOS, $DD = \$125.64$, so the ratio is $\sqrt{365} \times \$92.19 / \$125.64 = 14.0$.

A3. Maximum drawdown. $MDD = \min_t (E_t - \max_{s \leq t} E_s)$, where E_t is cumulative P&L, as a share of capital. v2 OOS: $-\$469 / \$22,754 = -2.06\%$.

A4. Probabilistic Sharpe. The chance that the true daily Sharpe beats SR^* , given the sample length T , skew γ_3 and kurtosis γ_4 (Bailey and López de Prado, 2014):

$$PSR(SR^*) = \Phi\left((\hat{SR} - SR^*)\sqrt{T-1} / \sqrt{1 - \gamma_3 \hat{SR} + \frac{\gamma_4 - 1}{4} \hat{SR}^2}\right).$$

v2 OOS: $\hat{SR} = 0.349$, $T = 40$, $\gamma_3 = 0.326$, $\gamma_4 = 3.26$, so $PSR(0) = 0.987$.

A5. Deflated Sharpe. $DSR = PSR(SR_0)$, where SR_0 is the best Sharpe we would expect from N tries on pure noise, with γ the Euler–Mascheroni constant:

$$SR_0 = \sqrt{V} \left((1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{Ne}\right) \right).$$

v2 OOS with $N = 3,386$ and $V = 1/(T-1)$: the two Φ^{-1} terms are 3.436 and 3.698, $SR_0 = 0.574$ a day (10.97 a year), and $DSR = 0.075$. Table 1 shows the lowest DSR over three choices of V . $N = 3,386$ is our variant count when the deflation ran; the count is now 4,219, which the haircut of Section 5 uses (at that N the OOS haircut is already 0).

A6. Probability of backtest overfitting. (Bailey et al., 2017). Split the daily P&L of 24 variants into $S = 16$ blocks. For each of the $\binom{16}{8} = 12,870$ ways to call half the blocks in sample, take the best in-sample variant, find its rank ω out of sample and set $\lambda = \ln \frac{\omega}{1-\omega}$; $PBO = \Pr(\lambda \leq 0)$. v2-safe grid: $PBO = 15\%$ (median $\lambda = 1.27$).

A7. Stationary block bootstrap. (Politis and Romano, 1994). Resample days in blocks of random length (mean 5 days) 10,000 times and read the percentiles. v2 OOS Sharpe: [1.9, 12.3], standard error 2.65, $P(SR \leq 0) = 0.0023$.

A8. Factor regression. $r_t = \alpha + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_U \text{Mom}_t + \varepsilon_t$ on daily return on capital, with Newey–West errors (5 lags) (Fama and French, 1993; Carhart, 1997; Newey and West,

1987; French, 2026). v2 IS (138 days): $\alpha = 0.801\%$ a day ($t = 8.48$); $\beta_M = 0.173$ ($t = 1.36$), $\beta_S = 0.085$, $\beta_H = 0.044$, $\beta_U = 0.028$; $R^2 = 3.3\%$.

A9. Taker fee. $f(q) = r q(1 - q)$ a share at price q , or $r(1 - q)$ per dollar, which is $10,000 r(1 - q)$ bps. At $r = 5\%$: $q = 0.5$ pays 1.25¢ (250 bps) and $q = 0.8$ pays 0.80¢ (100 bps).

A10. Point swing. $\Delta(k)$ of Section 2, with the match-win probability computed exactly by recursion over points, games, tiebreaks and sets (src/markov.py). Two equal players who win 0.645 of service points: at the first point $\Pr(\text{win}) = 50.0\%$, moving to 51.2% or 47.8%, so $\Delta = 3.4\%$. Serving at 5–5, 30–40 in the deciding set, $\Pr(\text{win}) = 33.7\%$, moving to 47.3% or 8.9%, so $\Delta = 38.4\%$. The mean $|\Delta|$ over a match is 5.7%.

A11. Latency budget. Eq. 1. With the pre-registered $L = 2.0$ s, $D = 1$ s and our measured $\delta = 0.119$ s, a point whose book reprices at the median $R = -1.16$ s needs $V < -0.28 + \ell$: no feed wins it without an early call. At the 90th percentile, $R = 0.08$ s and $V < 0.96$ s. The sweep's break-even (1.09 s IS) comes from these late reprices.

A12. Capital and return on capital. Capital is three times the peak dollars locked in open positions, each held 4 h or until resolution. v2 IS: peak \$9,434, capital \$28,302; return = $365 \bar{d}/\text{capital} = 365 \times \$196.24/\$28,302 = 253\%$ a year.

A13. Capacity and volume share. Capacity is the capital at which the seed-mean Sharpe falls to half its smallest-size value; share is our notional a day over market volume a day. CV, post hoc, IS, just below half Sharpe: \$10,428 a day against \$4.71M of in-play volume (0.22%) and \$117,047 traded by the fast tier 0–3 s after a point (9%).

Appendix B Everything we tested

Table A1. Every test we ran, its best result and our verdict. IS: in sample; OOS: out of sample (burned unless marked blind); ¢: net cents a share.

Test	What it was	Best result	Verdict
<i>Who is fast, and what they earn (public tapes)</i>			
H1 follow the jump	take the direction of a $\geq 4\text{¢}$ move, hold 30 s	OOS -2.08¢ $[-2.17, -1.98]$; 20 of 20 IS variants below zero	rejected
H2 favourites	buy in-play favourites priced 85–97¢	OOS $+0.80\text{¢}$ $[-1.36, 2.66]$	no edge
H4 score feeds	do public score feeds lag the book?	median lag 44.5 s ($n = 75$)	confirmed
H5 quote after jumps	post maker quotes after a move	OOS $+0.16\text{¢}$ $[-0.49, 0.87]$	not significant
H6 fast tier	wallets trading 0–3 s after a move, picked on past months	$+0.93 / +0.67\text{¢}$ net 30 s markout (IS / OOS prints only; $+0.75\text{¢}$ with August's IS prints); 9/9 and 3/3 months positive	confirmed
Kalshi lead-lag	Kalshi vs Polymarket on the same points	Kalshi first in 68.9% of 106,347 repricings (median 5 s)	lag mostly mechanical
Signal decay (TT5)	with-jump markout at 1–2 s minus 10–30 s after a move	OOS $+0.66$ $[0.45, 0.87]\text{¢}$; whole year $+0.19$ $[-0.26, 0.46]\text{¢}$; fast tier $+1.18$ to $+0.59\text{¢}$ (IS, 0 to 1–2 s)	decays in today's regime only
Clocks	when the book, the stamp and the tape move	book 1.2 s before the stamp ($n = 482$); tape 1.98 s late	measured
Factor regression	Fama–French three factors and momentum	$\alpha t = 8.48$; largest factor $ t = 1.36$	no factor exposure

Test	What it was	Best result	Verdict
<i>Copying the fast tier (v2 and relatives)</i>			
v1	copy the fast tier, dollar sizing; OOS opened once	IS +1.16¢; OOS +0.57¢, −\$36,056	failed
Sizing	55 sizing policies, walk-forward	G_50pct_net100: Sharpe 14.5 IS; all 55 with IS CI above zero (Sharpe 4.1–16.8)	adopted (v2); plateau
Exits	maker exit instead of holding (902 variants)	IS +0.53¢ [0.42, 0.64] (1 s / 5% regime); live fills adverse (27% filled in 30 s)	not adopted
Trade selection	which fast-tier trades to take (341 variants)	nested walk-forward +1.76¢ [1.36, 2.16] vs +1.16¢ (IS)	IS only
Side markets	side markets vs the match market (68 variants)	leaning maker +2.73¢ [1.73, 3.70] (IS); taking stale side quotes: no edge	IS only; tiny
v2	frozen copy book at the fast tier's own prices	Sharpe 14.5 / 6.7; OOS +0.60¢ [0.09, 1.13]	headline (Table 1)
v2, worse entry	enter ½ or 1 tick worse	IS +0.88 / +0.38¢; OOS +0.10 / −0.40¢	about zero OOS
v2, costs doubled	fees ×2 / all costs ×2	IS +0.77 / +0.27¢; OOS −0.34 / −0.84¢	fails OOS
v2, fixed costs	after central data and running costs	IS +\$29 a day; OOS −\$75 a day	fails OOS
v2, unseen markets	11,307 markets we never examined (blind)	IS +2.02¢; OOS +1.22¢ [−0.19, 2.65]	fail
v2-safe	net cap 50, fixed after v2's OOS losses	Sharpe 15.7 / 9.3; blind +1.21¢ [−0.02, 2.43]	fail (blind)
Per-wallet cap	\$1,000 a wallet a day plus retirement, chosen on IS from 12 variants, pre-registered	OOS +0.91¢ [0.20, 1.66], Sharpe 7.6, \$3,615 vs \$3,688; max DD −\$555 vs −\$469; top five 129% (was 142%)	pass (burned OOS); a loss limit, backtest only
Copier fill stress	v2 filled after the venue hold and the measured block lag, on the real tape	next same-side print: IS −0.47¢, OOS −1.37¢; optimistic: IS +0.67¢, OOS +0.21¢ [−0.32, 0.73]	a copier loses
Maker v1	pre-registered maker book (blind)	+1.87¢ [−0.14, 3.86] per fill, −\$379	fail
Rigor pack	deflated Sharpe, PBO, block bootstrap	DSR 0.997 / 0.075; PBO 15% (9–24% by block choice); $P(SR \leq 0) = 0.0023$	luck not ruled out OOS
PSR, MinTRL, haircut	six tests incl. a Bonferroni haircut over 4,219 variants	v2 IS 6 of 6 (haircut Sharpe 13.4); v2 OOS 3 of 6 ($t = 2.21$ vs 4.38); CV pre-registered 0, post hoc 4 OOS	no book passes all six OOS
Edge persistence	monthly decay, fee vs entrants, venue-rule breaks	net 0.16¢ a month down, before-fee −0.02¢ [−0.12, 0.07]; line hits zero in 3.3 months (0.6–12.8), a May step fits as well	cannot tell
Blind forward test	v2 on new matches, CV rule v3 as secondary	pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5)	not run
<i>Being fast ourselves (computer vision)</i>			

Test	What it was	Best result	Verdict
H3 early call	table-tennis MISS call 50 ms before contact, precision ≥ 0.95	live causal engine: 4 of 41 misses, none wrong; offline evaluation (look-ahead feature): 11/11	precision passes; recall low
Spin model, table tennis	spin-aware early call, held-out footage	10/12 right at 50 ms vs the frozen model's 11/11 (both offline)	not adopted
Tennis tracker, real clip	public TrackNet detector with our linking and court-line fit on a Pexels rally (no labels)	ball in 89% of 300 frames; 48/48 spot-checks on the ball; court lines 0.36 px (median)	tracks; no in/out calls
Broadcast tennis	single-camera landing and out calls, TrackNet games 8–10 (tuned on 1–7)	out calls 5 of 5 at the bounce, 3 of 3 at 33 ms, 3 of 4 at 67 ms; landing 17 cm at the bounce, 0.7–1.2 m at 33–300 ms	bounce only at 95%; detector saw all games
Tennis spin model	spin-aware landing model (simulated physics)	0.57 vs 5.76 cm at 200 ms; OUT precision 1, recall 0.98	simulation only
Rally gate	a miss call trades only within 2.0 s of a bounce call (engine log; 0.5–1.2 s also run)	removes 5 of 7 calls on unlabelled balls; keeps 3 of 5 right ones (2 at 0.5–0.8 s)	not enough
GPU engine	streamed causal engine on one L4	119.9 fps; 4.6 ms p50; 0 frames dropped	measured
WebRTC pipeline	our clip over WebRTC into the engine	46 ms p50, 67 ms p99 frame to call; video lag 4–8 ms	measured
End to end	video frame to unsigned order on a live book	54 ms ours; 2,119 ms with a 1 s feed vs 3,000 ms	pass (paper)
Courtside camera	CV trader with a camera at the venue ($V = 0$)	+\$89 / +\$46 a day (pre-registered)	simulated; no camera
Latency sweep	V from 0 to 60 s, six stamp-lag readings (944 cells)	pre-registered break-even 1.01–1.09 s	Table 1
CV trader, fees doubled	each fill pays its fee twice (Table 1)	every OOS cell loses (best –\$13 a day); fee 163–226 bps of notional	fails OOS
Fresh holdout, 0.5 s feed	Table-2 trader on 16 newer matches, pre-registered at dc95717	–\$20.5 (pre-registered) and +\$50.8 (post hoc) a day; CIs [–\$96, +\$57], [–\$38, +\$141]	anecdotal (2 days)
Price impact	log-size cost from 8.4 million IS taker orders; live-book walk	v2 1× OOS \$88 a day (conservative \$71, CI [–\$7, \$150]); others took 4–11% of stale depth	capacity holds; P&L thinner
CV rule v3	rule tuned on 360 IS variants	IS –0.22¢; blind –0.52¢	fail
Replay, all points	9 matches' real books, every point called	–0.94¢ [–1.28, –0.58] at $V = 1$ s; 36 of 36 settings below zero	loses
Replay, big points	only points with a large Markov swing (exploratory)	–0.95¢ [–1.64, –0.40]; 9 of 9 cells below zero	loses

Test	What it was	Best result	Verdict
Capacity	size grid for the CV book and v2	post hoc Sharpe halves at \$40,000–\$73,000; pre-registered: none	small
Table-tennis markets	TT1–TT5 on Polymarket table tennis	no fast wallet qualifies; spread 94¢	untestable

Appendix C Robustness and extra figures

These figures back the pipeline, regime, replay, latency and capacity claims in the main text, after the list of data sources (Table A2).

Table A2. *Real data behind every result; simulated: only CV call timing and the Hawk-Eye-class model.*

Source	What is real	Matches	Period	Used for
Polymarket books and trades (Polymarket, 2026a)	prices, fills, fees, delays	13,084 (8,554 ATP, 4,530 WTA); 1,310 at Slams (AO 415, RG 412, USO 483)	Oct 2025 – Oct 2026	fast tier; v2 (traded in 7,639 IS, 2,003 OOS); CV trader (7,755 matches at a 1 s hold; about 681 IS and 246 OOS traded a seed)
held back newest matches	same same	11,307 (10,855 ITF) 16 of 22	same Oct 3–4, 2026	blind test fresh holdout
Match Charting Project (Sackmann, 2026)	901,233 charted points	5,889 (3,337 men, 2,552 women)	2020–26	how points end (out 36–41%, net 25–26%)
TrackNet broadcast set (Huang et al., 2019)	19,835 labelled frames	10 (95 clips)		court, landing and out-call models
OpenTTGames (Voeikov et al., 2020); Pexels	120 fps video; phone video	41 held-out misses; one rally		early-call engine; tracker end to end

Frame to executable order in 2.12 s; our part 54 ms

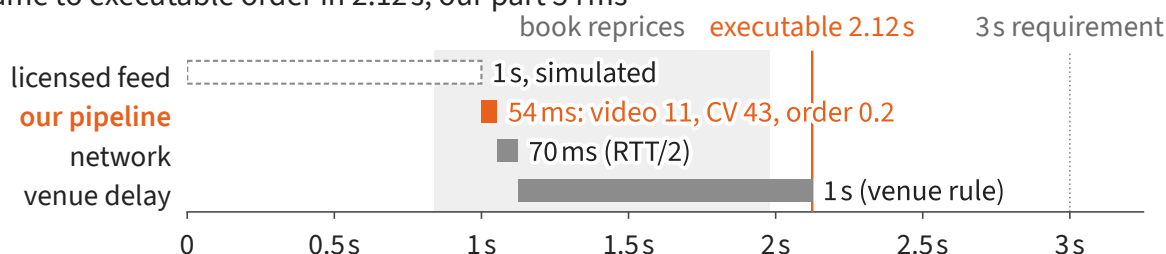


Figure A1. *We can trade inside 3 s. Mean stage times over 24 runs on our own footage; the 1 s feed is simulated; order built, not sent. Shaded: when the book reprices. Source: results/e2e.*

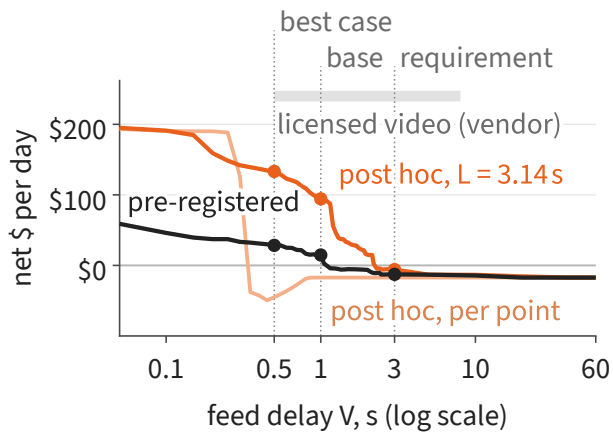
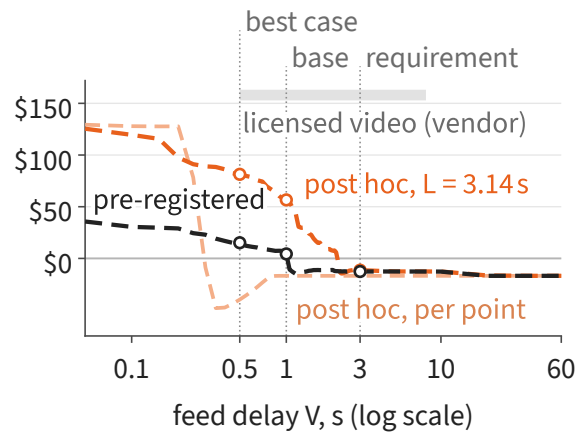
a IS: zero profit at 1.1–2.2 s of delay**b** OOS: zero profit at 1.0–2.1 s of delay

Figure A2. Each second of feed delay costs money. Simulated CV strategy, (a) IS, (b) burned OOS; dots: Table 1; grey bar: vendor-stated licensed video. Assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0.

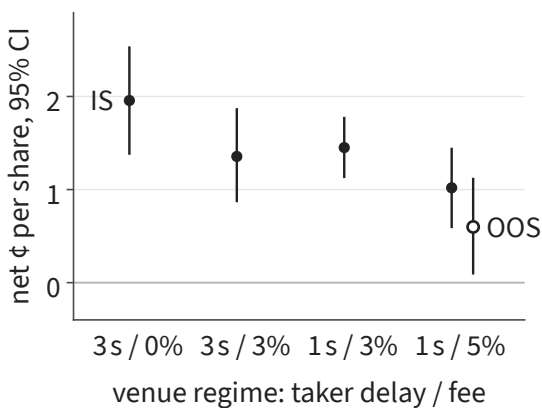
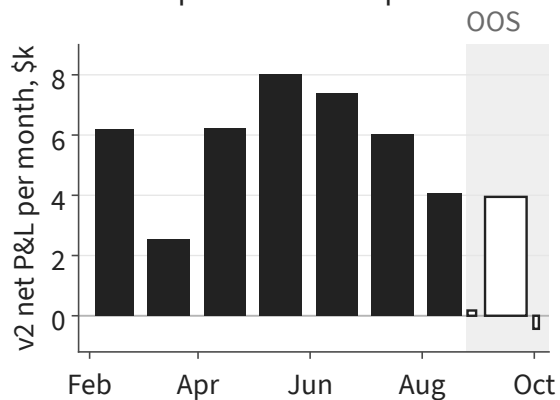
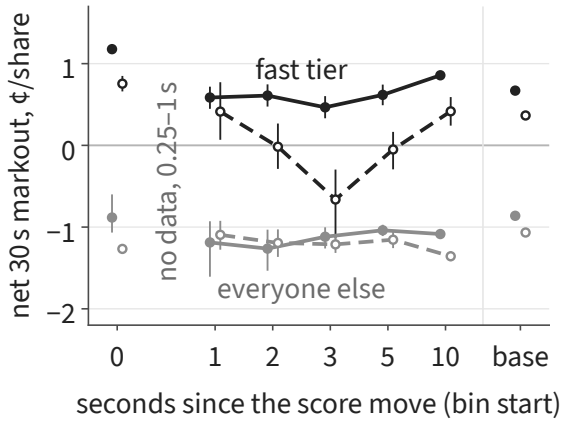
a Edge stays positive in every venue regime**b** 7 of 7 in-sample months are positive

Figure A3. v2's edge shrinks as the venue's protection shrinks, and stays positive in every regime. (a) Net \$ a share by venue regime (taker hold / fee), 95% CIs; (b) net P&L by month, at the fast tier's own prices. Source: results/risk/risk_stats.json.

a The fast tier stays ahead at every horizon



b Trading every point loses at every delay

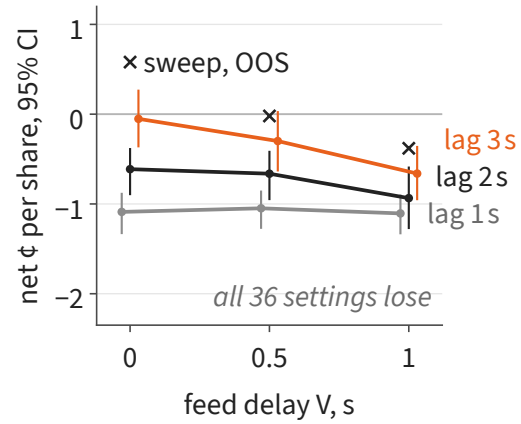
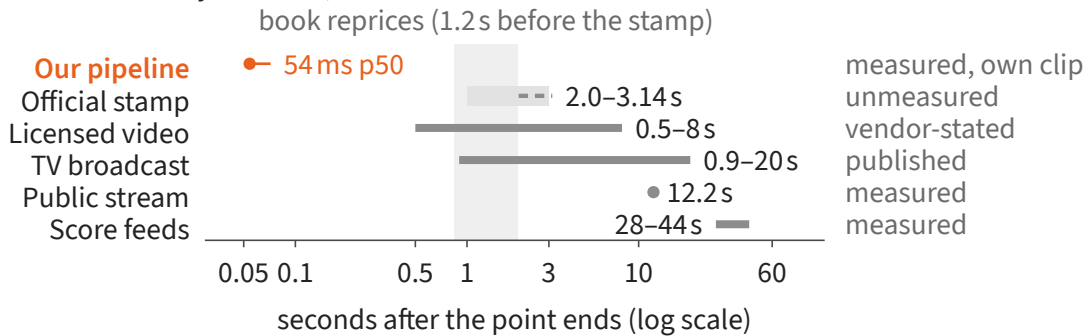


Figure A4. *The fast tier leads at every horizon; trading every point on real books loses.* (a) Net 30 s markout by seconds since the score move; filled: IS, hollow: OOS. (b) Replay of 9 matches recorded live on 2026-10-03 against their real order books, every point called; net ¢ a share marked at +30 s with match-clustered 95% CIs; x: the sweep's OOS cell. One day, assumed feed latency, paper only. Source: results/{decay,replay}.

a Our order is ready in 54 ms; feeds take seconds



b Live engine: 4 of 41 misses called early, none wrong

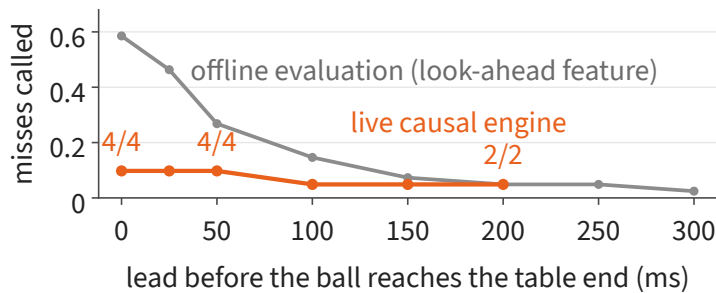
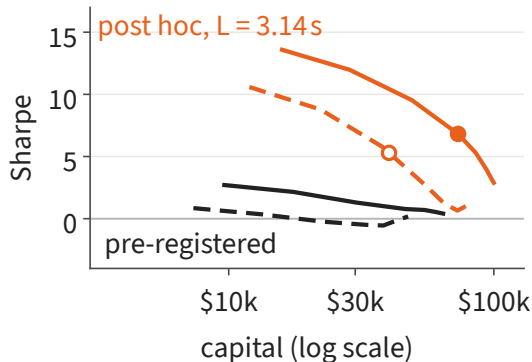


Figure A5. *Who sees the point first.* (a) Delay after the point ends (log scale): our pipeline (dot p50, whisker p99, own clip over WebRTC, laptop); the official stamp, dashed from the pre-registered 2.0s to the post hoc 3.14s over the cited 1–3s band (Regen Sports, 2026; Stats Perform, nd); vendor-stated licensed video (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Ably, nd); published stream delays (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026); measured public stream and score feeds. Shaded: when the book reprises. (b) Early out-calls on 41 held-out table-tennis misses: the live causal engine (orange), which we can deploy, and the offline evaluation (grey), which used a look-ahead feature. Source: results/{e2e,webrtc,tracking,engine}, research/v2/latency.

a Sharpe halves by \$40k–\$73k (post hoc)



b The cheapest data stack needs \$14k–\$16k

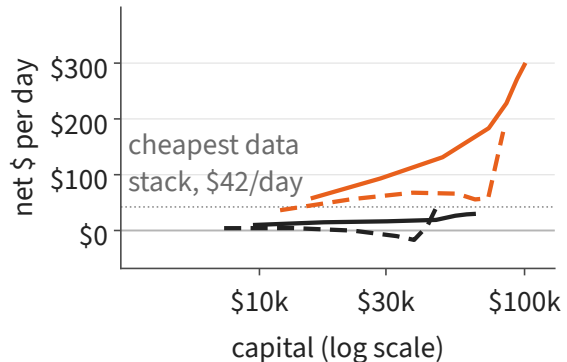


Figure A6. *How much capital the CV strategy can use at a 1 s feed.* (a) Sharpe and (b) net \$ a day against capital (net cap 50 to 5,000 shares, \$250 orders, 10 matches a day, half the stale depth); circles: where the post hoc Sharpe halves; dotted: the cheapest data stack. Solid: IS; dashed: burned OOS. Simulated, assumed feed latency (licensed feed not purchased); parameters measured. Source: results/capacity.

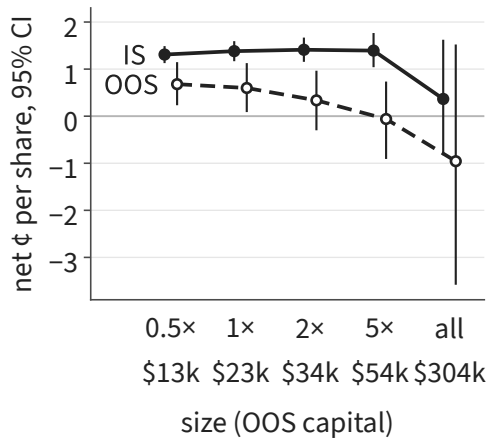
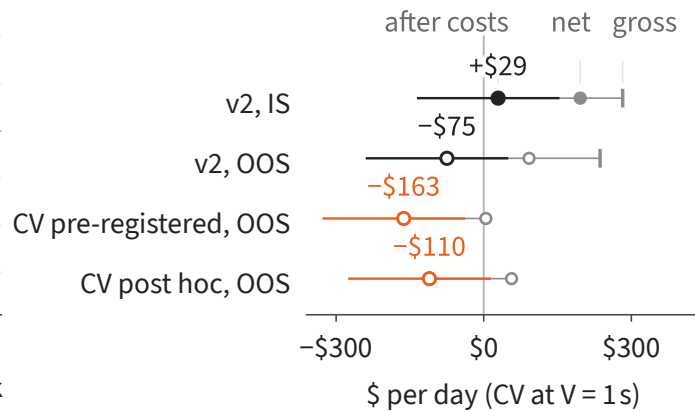
a v2's OOS edge fades as size grows**b** After fixed costs only v2 IS stays positive

Figure A7. v2's capacity is small and fixed costs bind. (a) Net edge by size (caps scaled; optimistic at large size, since we model no price impact); filled: IS, hollow: OOS. (b) \$ a day gross, after fees and after central fixed costs (bar: low to high); CV rows at the 1s base case, burned OOS, pre-registered first; assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

Appendix D Records

We keep the full logs in the repo instead of printing them here.

Pre-registration. H1-H4: HYPOTHESIS.md, committed at 7232986 (2026-10-03 09:50 UTC) before any result. H6, v2 and the forward test: HYPOTHESIS_V2.md (amendments A1-A5). Table tennis: HYPOTHESIS_TT.md. The CV simulation's research/v2/tier0/PREREG.md says it was written at 17:25 UTC; it was committed with its results at 19:24 UTC (a5769c7), and the peek log shows the OOS grid first ran at 17:29 UTC. git log on these files lists 19 commits.

Deviations. DEVIATIONS.md and research/v2/*/DEVIATIONS*.md give every rule change and its reason: 2 changes after a look for v2 and 13 for the CV simulation (T1-T3, V1-V10).

Reads of held-out data. results/oos_peeks.log has 111 lines, one per read: 7 blind first runs, 22 non-blind reads, 6 descriptive checks, 48 audits or re-runs, 22 live-data or replay reads and 6 others; 7 were written after their run and say so.

Variants. 4,219 in all (results/paper/variants.json): H1-H6 pre-registered and H5/H6 rules (44); v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi) (3,342); v2-safe grid (24); tier-0 pre-registered scenarios (432); tier-0 v3 in-sample grid (360); maker v1 evaluated variants (8); cost-stress cases (4); table-tennis tests TT1-TT5 (5). The fresh holdout, the fees-doubled CV cells and the PSR tests choose nothing and add no variant. Sensitivities that chose nothing: 944 latency-sweep cells and 36 replay cells, plus 4 rally-gate values (Section 4). The per-wallet cap of Section 6 added 12 more variants after this count. Two presentation choices were post hoc: the 3.14s headline reading and the 1s base case.

Grids and sources. Latency sweep: results/tier0/latency_sweep.json; replay: results/replay/selective/selective.json; fixed costs: results/financials/financials.json; latency sources: research/v2/latency/results.json; every number with its source: results/paper/numbers.json.

Forward test and live session. Blind forward test: pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5). The live paper session was stopped by a team decision and is not used (research/v2/maker/DEVIATIONS_LIVE.md, L15).